

Total marks: 20. Solve all the problems. Show your detailed work to receive the full marks.

1. Determine whether the function $y = 3x^3 \sin x - 9x^2 + 3$ is an odd, even or neither. [4]

Solution : Given,

$$y = 3x^3 \sin x - 9x^2 + 3$$

$$\Rightarrow f(x) = 3x^3 \sin x - 9x^2 + 3$$

$$\Rightarrow f(-x) = 3(-x)^3 \sin(-x) - 9(-x)^2 + 3$$

$$\Rightarrow f(-x) = -3x^3 \sin x - 9x^2 + 3$$

$$\Rightarrow f(-x) = 3x^3 \sin x - 9x^2 + 3$$

$$\Rightarrow f(-x) = f(x) \quad \therefore f(-x) = f(x)$$

$\therefore$ The given function is an even function. (Ans)

2. Find the limit using Squeeze theorem $\lim_{x \rightarrow -3} |x+3| \cos\left(\frac{1}{x+3}\right)$. [8]

Solution : We know,

$$-1 \leq \cos x \leq 1$$

$$\Rightarrow -1 \leq \cos\left(\frac{1}{x+3}\right) \leq 1$$

$$\Rightarrow -|x+3| \leq |x+3| \cos\left(\frac{1}{x+3}\right) \leq |x+3| \quad \left[\text{multiplying all sides with } |x+3| \right]$$

Applying Squeeze theorem,

$$\lim_{x \rightarrow -3} -|x+3| = 0$$

$$\lim_{x \rightarrow -3} |x+3| = 0$$

$$\therefore \lim_{x \rightarrow -3} f(x) = \lim_{x \rightarrow -3} g(x) = 0$$

$$\lim_{x \rightarrow -3} |x+3| \cos\left(\frac{1}{x+3}\right) = 0$$

(Ans)

3. Find the limit of $f(x)$ at $x = 3$, where

[8]

$$f(x) = \begin{cases} x^3 - 1, & 0 \leq x < 3 \\ 2 - 2x, & x = 3 \\ \frac{1}{x-3}, & x > 3. \end{cases}$$

Solution: For the given function,

$$\text{L.H.L} = \lim_{x \rightarrow 3^-} f(x)$$

$$= \lim_{h \rightarrow 0} f(3-h)$$

$$= \lim_{h \rightarrow 0} (3-h)^3 - 1$$

$$= 27 - 1$$

$$= 26$$

$$\text{R.H.L} = \lim_{x \rightarrow 3^+} f(x)$$

$$= \lim_{h \rightarrow 0} f(3+h)$$

$$= \lim_{h \rightarrow 0} \frac{1}{3+h-3}$$

$$= \lim_{h \rightarrow 0} \frac{1}{h}$$

$$= \infty$$

$$\therefore \text{L.H.L} \neq \text{R.H.L}$$

$\therefore$ Limit of $f(x)$ at $x = 3$ does not exist.

(Ans)